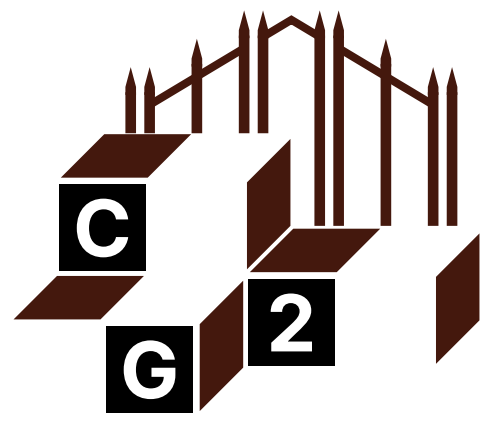
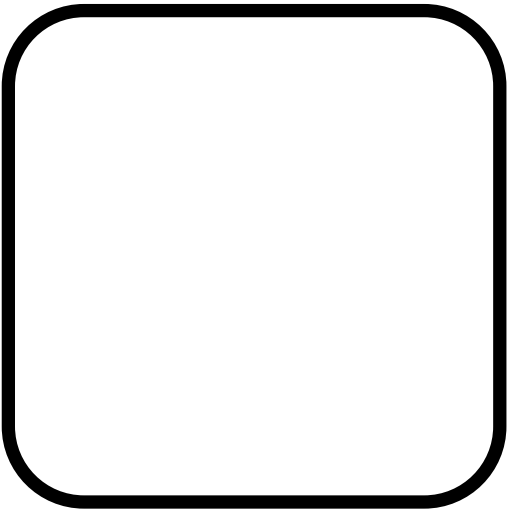




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Auslander-Reiten annihilators

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Introduction

The Auslander-Reiten Conjecture for commutative Noetherian rings predicts that a finitely generated module is projective when certain Ext-modules vanish. But what if those Ext-modules do not vanish? We study the annihilators of these Ext-modules and formulate a generalisation of the Auslander-Reiten Conjecture.

The Auslander-Reiten Conjecture

- Let R be a commutative Noetherian local ring and M a finitely generated R -module. We learn in a first course in homological algebra that M is a projective (=free) module if and only if for any finitely generated module X and for any $i \geq 0$ one has $\text{Ext}_R^i(M, X) = 0$.
- The Auslander-Reiten Conjecture predicts that instead of checking all finitely generated modules, it is enough to use $X = M \oplus R$.
- In other words, the Auslander-Reiten Conjecture states that if $\text{Ext}_R^i(M, M \oplus R) = 0$ for all $i \geq 1$, then M has to be a free module.

Auslander-Reiten annihilators

Motivated by the Auslander-Reiten Conjecture, we consider the ideal

$$\text{ARann}_R(M) := \bigcap_{i \geq 1} \text{ann}_R \text{Ext}_R^i(M, M \oplus R).$$

which we call the *Auslander-Reiten annihilator* of M . With this notation, the Auslander-Reiten Conjecture can be written as

$$M \text{ is free} \iff \text{ARann}(M) = R. \quad (1)$$

Stable annihilators

On the other hand, we consider another ideal

$$\underline{\text{ann}}_R(M) := \bigcap_{i \geq 1} \bigcap_{X \in \text{mod}(R)} \text{ann}_R \text{Ext}_R^i(M, X).$$

As mentioned above, it is well-known that

$$M \text{ is free} \iff \underline{\text{ann}}_R(M) = R. \quad (2)$$

We call $\underline{\text{ann}}_R(M)$ the *stable annihilator* of M as it is equal to the annihilator of the stable endomorphism ring of M . We always have

$$\underline{\text{ann}}_R(M) \subseteq \text{ARann}_R(M)$$

Our question

- Combining (1) and (2), we can rewrite the Auslander-Reiten Conjecture as

$$\underline{\text{ann}}_R(M) = R \iff \text{ARann}(M) = R.$$

- That is, for free modules, and only for free modules, the stable annihilator and the Auslander-Reiten annihilator agree and are both equal to R .
- What happens if we forget about the “and are both equal to R ” part?
- So, our main question is: for an arbitrary finitely generated R -module M , do we have an equality

$$\underline{\text{ann}}_R(M) = \text{ARann}(M)?$$

Example: Hypersurfaces

Let $R = k[[x_1, \dots, x_n]]/(f)$ be a hypersurface ring and M a maximal Cohen-Macaulay module. By the theory Tate cohomology of maximal Cohen-Macaulay modules over a Gorenstein ring and by the fact that one has 2-periodic resolutions over a hypersurface ring, we have an equality

$$\underline{\text{ann}}_R(M) = \text{ann}_R \text{Ext}_R^2(M, M)$$

and this tells us that our question has an affirmative answer for maximal Cohen-Macaulay modules over a hypersurface rings because one, by definition, has an inclusion

$$\text{ARann}_R(M) \subseteq \text{ann}_R \text{Ext}_R^2(M, M).$$

Theorem: Gorenstein isolated singularities

- We can prove the following theorem: Let R be a Gorenstein local ring which is an isolated singularity of Krull dimension at least 2. Then, our main question has an affirmative answer for every maximal Cohen-Macaulay module.
- Proof utilises Auslander-Reiten duality for isolated singularities and the following equality

$$\underline{\text{ann}}_R(M) = \text{ann}_R \text{Ext}_R^{d+1}(\text{Tr}(M), DM)$$

which works for all Cohen-Macaulay local rings with isolated singularities possessing a canonical module ω . Here, d is the Krull dimension, Tr is the Auslander transpose and $D = \text{Hom}_R(-, \omega)$ is the canonical dual.

A counterexample

Our question does not always have a positive answer. Lyle and Maitra found a counterexample where R is a numerical semigroup ring and M is a canonical module.

Further on Auslander-Reiten annihilators

For more on Auslander-Reiten annihilators, check

<https://www.sntp.ca/ozgur/arann>

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